# Tables

| Device | Architecture |
|---|---|
| D1 | FTO / PCBM / $RbGeI_3$ / NiO / Au |
| D2 | FTO / $TiO_2$ / $RbGeI_3$ / NiO / Au |
| D3 | FTO / PCBM / $RbGeI_3$ / CuI / Au |
| D4 | FTO / $TiO_2$ / $RbGeI_3$ / CuI / Au |
| D5 | FTO / PCBM / $RbGeI_3$ / CBTS / Au |
| D6 | FTO / $TiO_2$ / $RbGeI_3$ / CBTS / Au |
| D7 | FTO / PCBM / $RbGeI_3$ / Spiro-OMeTAD / Au |
| D8 | FTO / $TiO_2$ / $RbGeI_3$ / Spiro-OMeTAD / Au |

Table 1: Eight device configurations screened in this study. All configurations were simulated at an absorber thickness of 400 nm.

| Dev. | Architecture | $V_{OC}$ (V) | $J_{SC}$ ($mA/cm^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|---|
| D1 | FTO/PCBM/$RbGeI_3$/NiO/ Au | 0.81 | 30.10 | 83.68 | 20.40 |
| D2 | FTO/$TiO_2$/$RbGeI_3$/NiO/Au | 0.82 | 30.46 | 84.48 | 21.10 |
| D3 | FTO/PCBM/$RbGeI_3$/CuI/ Au | 0.81 | 30.10 | 83.96 | 20.47 |
| D4 | FTO/$TiO_2$/$RbGeI_3$/CuI/Au | 0.81 | 30.46 | 79.98 | 19.78 |
| D5 | FTO/PCBM/ $RbGeI_3$/CBTS/Au | 0.82 | 30.20 | 82.36 | 20.40 |
| D6 | FTO/$TiO_2$/$RbGeI_3$/CBTS/ Au | 0.82 | 30.55 | 82.13 | 20.57 |
| D7 | FTO/PCBM/ $RbGeI_3$/Spiro/Au | 0.81 | 30.07 | 85.01 | 20.71 |
| D8 | FTO/ $TiO_2$/$RbGeI_3$/Spiro/Au | 0.82 | 30.46 | 83.95 | 20.97 |

Table 2: Initial photovoltaic performance of the eight screened device configurations, simulated at an absorber thickness of 400 nm. Selected as the baseline for full parametric optimization.

| Step | Sweep range | Adopted | Gate basis |
|---|---|---|---|
| 1 thickness | 0.3–1.0 $\mu$m | 0.7 $\mu$m | sweep max, plateau |
| 2 bulk $N_t$ | 1e12–1e18 cm$^{-3}$ | 1e14 | passivated-film pin (max at 1e12 shown) |
| 3 $\epsilon_r$ | 10–35 | 15 | sweep-flat pin |
| 4 $\chi$ | 3.7–4.2 eV | 3.9 eV | interior max (3.7 non-convergent) |
| 5/6 interface $N_t$ | 1e12–1e18 cm$^{-2}$ | 1e14 | passivated-film pin (max at 1e12 shown) |
| 7 $E_g$ | 1.2–1.7 eV | 1.4 eV | SQ-margin gate (max at 1.3 rejected) |
| 8 $TiO_2$ | 10–90 nm | 10 nm | sweep max + pinhole caveat |
| 9 CuI | 0.05–1.9 $\mu$m | 100 nm | sweep-flat pin |
| 10/11 $N_C/N_V$ | 1e18–1e20 cm$^{-3}$ | 1e18 | literature floor (sub-floor $\leq$0.13 pp) |

Table 3: Per-step sweeps, adopted values and gate basis. Full traces: Figs. S1–S10.

| Parameter | $TiO_2$ | FTO | $RbGeI_3$ | CuI |
|---|---|---|---|---|
| Thickness (nm) | 10 | 500 | 700 | 100 |
| Bandgap $E_g$ (eV) | 3.2 | 3.2 | 1.4 | 3.1 |
| Electron affinity $\chi$ (eV) | 4.0 | 4.4 | 3.9 | 2.1 |
| Dielectric permittivity $\epsilon$r | 9 | 9 | 15 | 6.5 |
| CB effective DOS $N_C$ (cm$^{-3}$) | $2\times10^{18}$ | $2.2\times10^{18}$ | $1\times10^{18}$ | $2.8\times10^{19}$ |
| VB effective DOS $N_V$ (cm$^{-3}$) | $1.8\times10^{19}$ | $1.8\times10^{19}$ | $1\times10^{18}$ | $1\times10^{19}$ |
| Electron mobility $\mu$e (cm$^2$ V$^{-1}$ s$^{-1}$) | 20 | 20 | 28.6 | 100 |
| Hole mobility $\mu$p (cm$^2$ V$^{-1}$ s$^{-1}$) | 10 | 10 | 27.3 | 43.9 |
| Donor density $N_D$ (cm$^{-3}$) | $9\times10^{16}$ | $1\times10^{19}$ | $1\times10^{9}$ | 0 |
| Acceptor density $N_A$ (cm$^{-3}$) | 0 | 0 | $1\times10^{9}$ | $1\times10^{18}$ |
| Bulk defect density $N_t$ (cm$^{-3}$) | $1\times10^{15}$ | $1\times10^{14}$ | $1\times10^{14}$ | $1\times10^{14}$ |

Table 4: Optimized material parameters of the proposed FTO/$TiO_2$/$RbGeI_3$/CuI/Au perovskite solar cell. Transport-layer bulk defect densities were carried from the initial deck (not among the eleven swept steps).

| Device | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| Adopted reference (Table IV) | 1.06 | 30.32 | 84.11 | 27.06 |
| GA best-found ($E_g$ 1.36 eV) | 1.04 | 31.55 | 85.87 | 28.30 |
| Reference, $N_C$=$N_V$=1e19 | 0.97 | 30.32 | 84.09 | 24.82 |
| Reference + realistic optics (0.9 sun) | 1.06 | 27.28 | 83.97 | 24.28 |

Table 5: Adopted reference device versus GA best-found point and sensitivity rows (defect rows re-simulated, Fig. 8; optics row reports $PCE_{1-sun}$). PCEs are unrounded SCAPS outputs (displayed inputs rounded).

| Device Structure | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | FF (%) | PCE (%) | Year [Ref.] |
|---|---|---|---|---|---|
| FTO/$TiO_2$/$RbGeI_3$/NiO/Ag | 0.5311 | 28.89 | 63.68 | 9.77 | 2022 [**?**] |
| FTO/$TiO_2$/$CsGeI_3$/CuI/Au | 0.667 | 22.06 | 73.49 | 10.80 | 2022 [**?**] |
| ITO/$C_{60}$/$RbGeI_3$/CBTS/Ag | 0.99 | 33.20 | 82.80 | 27.21 | 2024 [**?**] |
| FTO/$TiO_2$/$RbGeI_3$/PEDOT:PSS/Au | 0.813 | 30.84 | 74.19 | 18.60 | 2025 [**?**] |
| FTO/$TiO_2$/$MASnI_3$/CBTS/Au | 1.030 | 30.24 | 83.41 | 25.98 | 2026 [**?**] |
| FTO/$TiO_2$/$MASnI_3$/CZTS/Au | 0.96 | 31.66 | 67.00 | 20.36 | 2025 [**?**] |
| FTO/$TiO_2$/$CsSnI_3$/P3HT/Au | 0.972 | 34.70 | 78.21 | 26.38 | 2024 [**?**] |
| FTO/$TiO_2$/$RbGeI_3$/Spiro-OMeTAD/Au | 1.1813 | 26.31 | 85.16 | 26.47 | 2026 [**?**] |
| FTO/$TiO_2$/$RbGeI_3$/CuI/Au (this work) | 1.06 | 30.32 | 84.11 | 27.06 | This work |
| FTO/$TiO_2$/$RbGeI_3$/$Cu_2O$-BSF/CuI/Au (this work) | 1.09 | 30.32 | 84.62 | 28.01 | This work |

Table 6: Comparison of the proposed device with recently reported lead-free perovskite solar cells. All comparator values verified against primary sources. All entries are SCAPS-1D simulation studies.

| Device | Condition | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | PCE (%) |
|---|---|---|---|---|
| $RbGeI_3$ | Ideal ($N_t$=1e14) | 1.06 | 30.32 | 27.06 |
| $CsGeI_3$ | Ideal ($N_t$=1e14) | 1.10 | 23.18 | 19.65 |
| $RbGeI_3$ | Realistic ($N_t$=1e15) | 1.02 | 30.31 | 24.40 |
| $CsGeI_3$ | Realistic ($N_t$=1e15) | 1.10 | 23.17 | 19.31 |
| $RbGeI_3$ | High oxidation ($N_t$=1e16) | 0.96 | 30.22 | 20.76 |
| $CsGeI_3$ | High oxidation ($N_t$=1e16) | 1.09 | 23.10 | 17.07 |

Table 7: $RbGeI_3$ versus $CsGeI_3$ in the identical FTO/$TiO_2$/absorber/CuI/Au stack under ideal, realistic, and high-oxidation defect budgets. Only the absorber bandgap differs (1.4 vs 1.6 eV) at fixed affinity — a gap-effect control, not A-site chemistry proof.

| Device | $V_{OC}$ (V) | $J_{SC}$ (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| Wang control (measured) | 1.05 | 20.03 | 72.0 | 15.10 |
| SCAPS rebuild + $R_s$ 6.9 | 1.060 | 19.92 | 71.88 | 15.18 |
| Wang champion (measured) | 1.11 | ∼20.0 | 82.0 | 18.40 |
| SCAPS rebuild | 1.127 | 19.92 | 82.37 | 18.49 |

Table 8: Calibration check: SCAPS rebuild vs measured $\beta$-$CsPbI_3$ devices [**?**] at 318 K. Fitted knobs are defect densities (plus $R_s$ on the control); the control fit misses the knee.